Exercise 1

In the lecture we convinced ourselves that all maps are well-defined. As an inclusion, the first map is clearly injective.

Exactness at $\mathcal{O}_k^\times$ is clear, since $\rho(u) = 1 \in (\mathcal{O}_k/\mathfrak{m})^\times$ by definition means exactly $u \equiv 1 \mod {}^\times\mathfrak{m}$.

Exactness at $(\mathcal{O}_k/\mathfrak{m})^\times$: For $u \in \mathcal{O}_k^\times$ we have $(u) = (1)$, so $(u)\mathcal{P}_k(\mathfrak{m}) = \mathcal{P}_k(\mathfrak{m})$. Conversely, let $\alpha \in k^\times$, $(\alpha, \mathfrak{m}_0) = 1$ and suppose $(\alpha)\mathcal{P}_k(\mathfrak{m}) = \mathcal{P}_k(\mathfrak{m})$. That is, there is some $\varepsilon \in \mathcal{O}_k^\times$ s.t. $\varepsilon\alpha \equiv 1 \bmod {}^\times\mathfrak{m}$, and $\rho(\alpha) = \rho(\varepsilon^{-1}) \in \rho(\mathcal{O}_k^\times)$.

Exactness at $\mathrm{cl}_k(\mathfrak{m})$ is clear, since both kernel and image consist exactly of the classes of principal ideals coprime to $\mathfrak{m}$.

Finally, we show more generally, that if $\mathfrak{m} \mid \mathfrak{n}$, then the induced map $\mathrm{cl}_k(\mathfrak{n}) \to \mathrm{cl}_k(\mathfrak{m})$ is surjective: Let $\mathfrak{a}\mathcal{P}_k(\mathfrak{m}) \in \mathrm{cl}_k(\mathfrak{m})$, for some ideal $\mathfrak{a} \in I_k(\mathfrak{m})$. By the approximation theorem, we find $\alpha \in k$ with $\alpha \equiv 1 \bmod {}^\times\mathfrak{m}$ and $\nu_\mathfrak{p}(\alpha) = -\nu_\mathfrak{p}(\mathfrak{a})$ for all prime ideals $\mathfrak{p} \mid (\mathfrak{n}, \mathfrak{a})$. Then $(\alpha\mathfrak{a}, \mathfrak{n}) = 1$, and

$$\alpha\mathfrak{a}\mathcal{P}_k(\mathfrak{n}) \mapsto \alpha\mathfrak{a}\mathcal{P}_k(\mathfrak{m}) = \mathfrak{a}\mathcal{P}_k(\mathfrak{m}).$$

Exercise 2

If $0 \to A \to B \to C \to 0$ is a short exact sequence of groups, then B is partitioned into $|C|$ cosets each of size $|A|$, hence $|B| = |A| \cdot |C|$. If $0 \to A \to B \to C \to D \to 0$ is exact, let $K = \ker(C \to D) = \mathrm{im}(A \to B)$. Then $0 \to A \to B \to K \to 0$ and $0 \to K \to C \to D \to 0$ are exact, and therefore

$$|A| \cdot |C| = |A| \cdot |K| \cdot |D| = |B| \cdot |D|.$$

(More generally, given any finite exact sequence, continuing in this way yields an equation of "alternating" cardinalities.)

Applying this to the exact sequence $0 \to \mathcal{O}_k^\times/(\mathcal{O}_k^\times \cap k_\mathfrak{m}^\times) \to \ldots \to 0$ yields

$$[\mathcal{O}_k^\times : \mathcal{O}_k^\times \cap k_\mathfrak{m}^\times] \cdot |\,\mathrm{cl}_k(\mathfrak{m})| = |\,\mathrm{cl}_k\,| \cdot |(\mathcal{O}_k/\mathfrak{m})^\times|.$$

In particular, since the right hand side is finite, and groups cannot be empty, both cardinalities on the left hand-side are finite as well, and dividing by the first term yields the result.

Exercise 3

Let $k = \mathbb{Q}(\sqrt{d})$ and let $\tau_1 = \mathrm{id}, \tau_2 : \bar{\cdot}$ be the two embeddings associated to $\infty_{1/2}$. By Dirichlet, $\mathcal{O}_K^\times = \{\pm 1\} \times \varepsilon^\mathbb{Z}$, where wlog $\varepsilon > 1$. Clearly $-1 \not\equiv 1 \bmod {}^\times\mathfrak{m}$. From $\mathrm{N}_{k/\mathbb{Q}}(\varepsilon) = \varepsilon\bar{\varepsilon}$ we see $\mathrm{sgn}(\bar{\varepsilon}) = \mathrm{N}_{k/\mathbb{Q}}(\varepsilon)$. Hence if $\mathrm{N}_{k/Q}(\varepsilon) = 1$, then $\mathcal{O}_k^\times \cap k_\mathfrak{m}^\times = \varepsilon^\mathbb{Z}$ and the index in $\mathcal{O}_k^\times$ is 2. Otherwise, $\mathcal{O}_k^\times \cap k_\mathfrak{m}^\times = \varepsilon^{2\mathbb{Z}}$ and the index in $\mathcal{O}_k^\times$ is 4.

Further, since $\mathfrak{m}_0 = (1)$ we have $|(\mathcal{O}_k/\mathfrak{m})^\times| = |\{\pm 1\}^2| = 4$. The result now follows from exercise 2.

Exercise 1

We have

$$\frac{(\mathcal{O}_K/\mathfrak{f})^\times}{U} \cong \ker(\mathrm{cl}_k(\mathfrak{f}) \to \mathrm{cl}_k) \cong \ker(\mathrm{Gal}(k(\mathfrak{f})/k) \to \mathrm{Gal}(k(1)/k)) \cong \mathrm{Gal}(k(\mathfrak{f})/k(1)),$$

where the first isomorphism follows from last week's sheet, the central one is induced by the Artin map (which is functorial), and the last is Galois theory.

Exercise 2

Let $\mathfrak{m} := q\mathbb{Z}\infty$. Then we know $\mathbb{Q}(\mathfrak{m}) = \mathbb{Q}(\zeta_q)$ and hence the Artin map induces an isomorphism

$$\mathrm{cl}_{\mathbb{Q}}(\mathfrak{m}) \cong \mathrm{Gal}(\mathbb{Q}(\zeta_q)/\mathbb{Q}) \cong (\mathbb{Z}/q\mathbb{Z})^\times \cong \mathbb{Z}/(q-1)\mathbb{Z}.$$

Let $H \subseteq \mathrm{cl}_Q(\mathfrak{m})$ be the unique subgroup of index 2. Then the existence theorem yields a unique abelian subfield $\mathbb{Q}(\zeta_q)/K/\mathbb{Q}$ with $\mathrm{Gal}(K/\mathbb{Q}) \cong \mathrm{cl}_{\mathbb{Q}}(\mathfrak{m})/H$, and ν ramifies in K only if $\nu \mid \mathfrak{m}$. Now the first condition implies that $K/\mathbb{Q}$ is quadratic, say $K = \mathbb{Q}(\sqrt{d})$ for some squarefree $d \in \mathbb{Z}$. Let p be a prime divisor of d. Then p ramifies in K, hence $p \mid q$. Hence either $d = -1$, or $d = \pm q$. Now 2 isn't allowed to ramify, so $d = -1$ is out, and we need the sign so that $\pm q \equiv 1 \bmod 4$. Hence if $q \equiv 1 \bmod 4$, we have $\mathbb{Q}(\sqrt{q}) \subseteq \mathbb{Q}(\zeta_q)$, and if $q \equiv 3 \bmod 4$, then $\mathbb{Q}(\sqrt{-q}) \subseteq \mathbb{Q}(\zeta_q)$, hence

$$\mathbb{Q}(\sqrt{q}) \subseteq \mathbb{Q}(\sqrt{-q}, \sqrt{-1}) = \mathbb{Q}(\sqrt{-q})\mathbb{Q}(\zeta_4) \subseteq \mathbb{Q}(\zeta_q)\mathbb{Q}(\zeta_4) = \mathbb{Q}(\zeta_{4q}).$$

In both cases this implies the claim, since $\mathbb{Q}(\sqrt{q})$ is totally real.

Exercise 3

Let $m := [k : \mathbb{Q}_p]$. Then we showed in ANT that $k^\times \cong \mathbb{Z} \oplus \mathbb{Z}/(q-1)\mathbb{Z} \oplus \mathbb{Z}/p^a\mathbb{Z} \oplus \mathbb{Z}_p^m$ for a p-power q and non-negative integer a. Hence

$$k^\times/(k^\times)^n \cong \mathbb{Z}/n\mathbb{Z} \oplus \mathbb{Z}/(n, q-1)\mathbb{Z} \oplus \mathbb{Z}/(n, p^a)\mathbb{Z} \oplus (\mathbb{Z}_p/n\mathbb{Z}_p)^m.$$

The first three summands are clearly finite, hence it remains to consider $\mathbb{Z}_p/n\mathbb{Z}_p$. Write $n = p^v b$ with $v, b \geq 0$, $p \nmid b$. Then b is invertible in $\mathbb{Z}_p$, hence $\mathbb{Z}_p/n\mathbb{Z}_p \cong \mathbb{Z}_p/p^v\mathbb{Z}_p$, which is isomorphic to $\mathbb{Z}/p^v\mathbb{Z}$ by ANT. Putting everything together, we see that $(k^\times)^n$ has finite index in $k^\times$.

To see that $(k^\times)^n$ is open, it suffices to find an $N \in \mathbb{N}$ with $1 + \mathfrak{p}_k^N \subseteq (k^\times)^n$, then $(k^\times)^n = \bigcup_{\alpha \in (k^\times)^n} \alpha(1 + \mathfrak{p}_k^N)$ is a union of open sets, hence open. To do this, let $s > \frac{e}{p-1}$ and $N = s + v_k(n)$. Then

$$1 + \mathfrak{p}_k^N = \exp(n\mathfrak{p}_k^s) = \exp(\mathfrak{p}_k^s)^n = (1 + \mathfrak{p}_k^s)^n \subseteq (k^\times)^n.$$

Exercise 4

Let K/k be finite abelian of exponent p. By local class field theory, for $a \in k^\times$ we have $(a^p, K/k) = (a, K/k)^p = \mathrm{id} \in \mathrm{Gal}(K/k)$. Hence $(k^\times)^p \subseteq \ker((\_, K/k))$. Conversely, if $(k^\times)^p \subseteq \ker((\_, K/k))$, then since the Artin map is surjective, $\mathrm{Gal}(K/k)$ has exponent (dividing) p. Hence the finite abelian extensions of exponent p correspond exactly to the subgroups $(k^\times)^p \subseteq H \subsetneq k^\times$ (they are automatically open).

By the same calculation as in exercise 3, for $p \neq 2$ we have $k^\times/(k^\times)^p \cong \mathbb{Z}/p\mathbb{Z} \oplus \mathbb{Z}/p\mathbb{Z}$. This has $\frac{p^2-1}{p-1} = p + 1$ subgroups of order p, and the trivial subgroup, so there are $p + 2$ finite abelian extensions of $\mathbb{Q}_p$ of exponent p. For $p = 2$, the same method yields $k^\times/(k^\times)^2 \cong (\mathbb{Z}/2\mathbb{Z})^3$, which has 7 subgroups of order 2, and $\frac{1}{3}\binom{7}{2} = 7$ subgroups of order 4, so there are $1 + 7 + 7 = 15$ finite abelian extensions of $\mathbb{Q}_2$ of exponent 2.

Exercise 1

Since $k^\times \subseteq J_k$ are Hausdorff topological groups, it suffices to show that $\{1\} \subseteq k^\times$ is open, i.e. that there exists an open neighbourhood $1 \in U \subseteq J_k$ with $U \cap k^\times = \{1\}$. Let S_∞ be the set of real places of k, and set $U_\varepsilon := \prod_{v \in S_\infty} B_\varepsilon(1) \times \prod_{v \notin S_\infty} \mathcal{U}_v$ for $\varepsilon > 0$. By definition of the product topology, U_ε is open. Now let $\alpha \in k^\times$. Then $\alpha \in U_\varepsilon$ implies $\alpha \in \mathcal{O}_{k_v}$ for the finite places, i.e. $\alpha \in \mathcal{O}_k$, and $|\alpha - 1|_v < \varepsilon$ for every infinite place $v \in S_\infty$. But $\mathcal{O}_k$ embeds as a lattice in $\mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$, which is in particular discrete. So for ε small enough, no $\alpha \in \mathcal{O}_k$ satisfies these conditions.

Exercise 2

Let cl_k be generated by the classes of $\mathfrak{p}_1, \dots, \mathfrak{p}_r$. Then set $S := S_\infty \cup \{\mathfrak{p}_1, \dots, \mathfrak{p}_r\}$. Let $\alpha \in J_k$ and denote by $\varphi : J_k/k^\times \mathcal{U}_k \to \mathrm{cl}_k$ the isomorphism from exercise 3. Then by construction, write $\varphi(\alpha) = x \prod \mathfrak{p}_i^{e_i}$ for some $e_i \in \mathbb{Z}$ and $x \in K^\times$. Set $\beta = \alpha x^{-1}$, we have to show $\beta \in J_k^S$. So let $v \notin S$, then v is finite. We have $v(\alpha) = v(\varphi(\alpha)) = v(x)$ since $v \notin \{\mathfrak{p}_1, \dots, \mathfrak{p}_r\}$, but $\varphi(\alpha)$ and x only differ by factors of the $\mathfrak{p}_i$. Hence $v(\beta) = v(\alpha) - v(x) = 0$ and $\beta \in \mathcal{U}_v$, as required.

From $J_k = J_k^S k^\times$ it immediately follows that $C_k^S = J_k^S k^\times / k^\times = J_k/k^\times = C_k$.

Exercise 3

(a) Consider the content map $\overline{c} : J_k \xrightarrow{c} I_k \twoheadrightarrow \mathrm{cl}_k$. The kernel of $\overline{c}$ is $c^{-1}(k^\times) = k^\times \ker c = k^\times \mathcal{U}_k$.

(b) Consider the commutative diagram defining the Artin map on J_k (after theorem 1.45).Let $\mathfrak{m} = 1$ and $K = k(1)$. We know that ideal-theoretically, K corresponds to $P_k(\mathfrak{m})$. Also, the top arrow is induced by the content, so it is exactly the map in (a). Finally, since $\mathfrak{m} = 1$ the left vertical arrow is the identity. Under all these simplifications, the Artin map reduces to the composition

$$J_k \twoheadrightarrow J_k/k^\times \mathcal{U}_k \xrightarrow[\cong]{\overline{c}} \mathrm{cl}_k \xrightarrow[\cong]{(\_, K/k)} \mathrm{Gal}(K/k).$$

Since the last two arrows are isomorphisms, the kernel is just the kernel of the first one, which is obviously $k^\times \mathcal{U}_k$.

Exercise 4

(a) It's clear that $\mathfrak{p}$ ramifies in L/k if and only if $\sigma(\mathfrak{p})$ ramifies in $\widehat{\sigma}(L)/k$. So $\widehat{\sigma}(L)/k$ is unramified outside $\sigma(\mathfrak{m})$, and the Artin map is well-defined on $I_k(\sigma(\mathfrak{m}))$. Now let $\mathfrak{p} \in I_k(\mathfrak{m})$ be a prime ideal, and $\tau := (\mathfrak{p}, L/k) \in \mathrm{Gal}(L/k)$ its Frobenius. Then τ is uniquely determined by $\tau(\alpha) \equiv \alpha^f \bmod \mathfrak{p}$, where $f := \mathrm{N}(\mathfrak{p})$. Let $\alpha \in \widehat{\sigma}(L)$, then $\tau(\widehat{\sigma}^{-1}\alpha) \equiv (\widehat{\sigma}^{-1}\alpha)^f \equiv \widehat{\sigma}^{-1}(\alpha^f) \bmod \mathfrak{p}$, hence $\widehat{\sigma}\tau\widehat{\sigma}^{-1}(\alpha) \equiv \widehat{\sigma}\widehat{\sigma}^{-1}(\alpha^f) \equiv \alpha^f \bmod \sigma(\mathfrak{p})$, which precisely means $(\sigma(\mathfrak{p}), \widehat{\sigma}(L)/k) = \widehat{\sigma}(\mathfrak{p}, L/K)\widehat{\sigma}^{-1}$. Since the prime ideals generate the fractional ideals, this identity automatically holds for all fractional ideals.

In particular, using that L is defined mod $\mathfrak{m}$, i.e. $P_k(\mathfrak{m}) \subseteq \ker(\_, L/k)$, we see for $\sigma(\mathfrak{a}) \in \sigma(P_k(\mathfrak{m})) = P_k(\sigma(\mathfrak{m}))$ that $(\sigma(\mathfrak{a}), \widehat{\sigma}(L)/k) = \widehat{\sigma}(\mathfrak{a}, L/k)\widehat{\sigma}^{-1} = \widehat{\sigma}\widehat{\sigma}^{-1} = 1$, i.e. $P_k(\sigma(\mathfrak{m})) \subseteq \ker(\_, \widehat{\sigma}(L)/k)$, as desired.

(b) Since $\ker(\_, \widehat{\sigma}(k_H)/k) = \sigma(\ker(\_, k_H/k))$, we see that $\widehat{\sigma}(k_H)$ corresponds to $\sigma(H)$.

Now k_H/F is galois if and only if $\widehat{\sigma}(k_H) = k_H$ for all $\widehat{\sigma}$, if and only if $\sigma(H) = H$, for all $\sigma \in G$, if and only if $\sigma(H) \subseteq H$ for all $\sigma \in G$.

Exercise 1

Let $\mathfrak{m} := \mathfrak{q}$ be a prime of k. By exercise 1.28, $[k(\mathfrak{m}) : H] = \frac{\omega(1)}{\omega(\mathfrak{m})}\varphi_k(\mathfrak{m}) = \frac{\omega(1)}{\omega(\mathfrak{m})}(\mathrm{N}(\mathfrak{q})-1)$. If $\mathfrak{q}$ is split in $k/\mathbb{Q}$, then $\mathrm{N}(\mathfrak{q}) =: q$ is a prime number, and by class field theory there exists a field $H \subseteq K \subseteq k(\mathfrak{m})$ with the desired properties (the conductor must divide $\mathfrak{q}$ and cannot be trivial) if and only if $p \mid [k(\mathfrak{m}) : H]$. Hence it suffices to show that there are infinitely many primes $q \equiv 1 \bmod p$ that split in k.

If $p = 2$ this is clear, so now let p be odd and $k = \mathbb{Q}(\sqrt{d})$, $d < 0$. q splits in k if and only if $1 = \left(\frac{d}{q}\right)$. Write $d = p^n e$ with $p \nmid e$, then by quadratic reciprocity, q splits in k if and only if $\left(\frac{e}{q}\right) = (-1)^{n(p-1)(q-1)/4}$, which is a congruence condition $\mathrm{mod} 4e$. By the Chinese Remainder Theorem and Dirichlet's Theorem on Arithmetic Progressions, there are infinitely many primes $q \equiv 1 \bmod p$ satisfying this condition, hence infinitely many field extensions as required.

Exercise 2

Clearly it suffices to prove (b). Recall that I_G is the free abelian group generated by $\sigma - 1$, $1 \neq \sigma \in G$. Hence set $\varphi : I_G \to G/G'$, $\sigma - 1 \mapsto \sigma$ and extend linearly. Now

$$\varphi((\sigma-1)(\tau-1)) = \varphi((\sigma\tau - 1) - (\sigma - 1) - (\tau - 1)) = \sigma\tau\sigma^{-1}\tau^{-1}G' = 1,$$

so $I_G^2 \subseteq \ker\varphi$ and we get a well-defined morphismn $\overline{\varphi} : I_G/I_G^2 \to G/G'$.

Conversely, let $\psi : G \to I_G/I_G^2$, $\sigma \mapsto \sigma - 1 + I_G^2$. For $\sigma, \tau \in G$, we have

$$\psi(\sigma\tau) = \sigma\tau - 1 + I_G^2 = (\sigma - 1) + (\tau - 1) + (\sigma - 1)(\tau - 1) = \psi(\sigma) + \psi(\tau).$$

Thus ψ is a group homomorphism. Since I_G/I_G^2 is abelian, it factors through $\overline{\psi} : G/G' \to I_G/I_G^2$. Clearly $\overline{\varphi}$ and $\overline{\psi}$ are mututally inverse, hence $I_G/I_G^2 \cong G/G' = G^{\mathrm{ab}}$.

Exercise 3

Let $G = \{\pm 1\}$, $A = \mathbb{Z}/2\mathbb{Z}$ with the trivial G-action and $B = \mathbb{Z}/4\mathbb{Z}$ with the action $-1 \cdot b = -b = 3b$. Then $B^G = \{0, 2\} \cong \mathbb{Z}/2\mathbb{Z}$, and $A \otimes B^G \cong (Z/2\mathbb{Z}) \otimes (\mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ (recall that, in general, multiplication induces an isomorphism $C_n \otimes C_m \cong C_{\gcd(n,m)}$) with the non-trivial element $1 \otimes 2$. Its image in $A \otimes B = (\mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z}/4\mathbb{Z})$ is $1 \otimes 2 = 2(1 \otimes 1) = 2 \otimes 1 = 0$, so the canonical map $A^G \otimes B^G \to (A \otimes B)^G$ is the zero map. Since $1 \otimes -1 = 3(1 \otimes 1) = 1 \otimes 1$, we have $(A \otimes B)^G = A \otimes B \cong \mathbb{Z}/2\mathbb{Z}$, so the map is not surjective either, even if the action of G on A is trivial.

Exercise 4

(1)⇒(4): Let $0 \to A \to B \to X \to 0$ be exact, with X projective. Thus there exists a morphism making the following diagram commute:

$$\begin{tikzcd} & X \arrow[dl, dashed, "\exists s"'] \arrow[d, "\mathrm{id}"] & \\ B \arrow[r] & X \arrow[r] & 0 \end{tikzcd}$$

which is exactly the required splitting.

(4)⇒(3): The morphism $\pi : B := \bigoplus_{x \in X} \mathbb{Z}[G] \to X$, $x \mapsto x$ is clearly surjective. By assumption let $s : X \to A$ be a splitting of the short exact sequence $0 \to \ker\pi \to B \to X \to 0$, then define $\varphi : \ker\pi \oplus X \to B$, $(a, x) \mapsto a + s(x)$. Let $(a, x) \in \ker\varphi$, then $0 = \pi(\varphi(a, x)) = \pi(\sigma(x)) = x$, hence also $a = \varphi(a, 0) = 0$. Conversely, let $b \in B$. Then $\pi(b - \sigma(\pi(b))) = 0$, so $b - \sigma(\pi(b)) = a$ for some $a \in \ker \pi$. But that exactly means $b = \varphi(a, \sigma(\pi(b)))$, so φ is an isomorphism and X is direct summand of a free G-module.

(3)⇒(2): Let $X \oplus Y = Z$ with Z a free G-Module. The result follows immediately from $\mathrm{Hom}(Z, \_)$ being exact and $\mathrm{Hom}(Z, \_) = \mathrm{Hom}(X, \_) \times \mathrm{Hom}(Y, \_)$.

(2)⇒(1): Let $g : B \to C$ be surjective, and $h : X \to C$ any morphism. Then by assumption $\mathrm{Hom}(X, B) \to \mathrm{Hom}(X, C) \to 0$ is surjective, so there is a preimage $s : X \to B$ of h.

Exercise 1

(a) Let $a \in A^G$. Then $gi(a) = i(ga) = i(a) \in B^G$, so the induced sequence is well-defined. It is clear that $i|_{A^G}$ is injective, since i is, and that $j|_{B^G} \circ i|_{A^G} = 0$. Conversely, let $b \in \ker j|_{B^G}$. Since the original sequence is exact, there is $a \in A$ with $i(a) = b$, we have to show $a \in A^G$. For $g \in G$, $i(ga) = gi(a) = gb = b = i(a)$ and the claim follows from the injectivity of i.

(b) By (a) and the long exact cohomology sequence, exactness everywhere but at C^G and $H^1(G, C)$ is clear, and the latter follows directly from $\operatorname{im} \delta_G = \operatorname{im} \delta_0$. For $b \in B^G$ we have $\delta_G(j(b)) = \delta_0(j_*(\bar{b})) = 0$. Conversely, let $c \in \ker \delta_G$, i.e. $\delta_0(\bar{c}) = 0$. By exactness of the long exact cohomology sequence, there is $b \in B^G$ such that $j_*(\bar{b}) = \bar{c}$. Hence $c = j(b) + \mathrm{N}_G\, c'$ for some $c' \in C$. Let $b' \in B$ be a preimage of c', then $c = jb + \mathrm{N}_G\, jb' = j(b + \mathrm{N}_G\, b') \in \operatorname{im} j|_{B^G}$.

Exercise 2

Let $L = \mathbb{Q}(\sqrt{d})$ with $0, 1 \neq d \in \mathbb{Z}$ squarefree. Let $h = 1$ if $d \equiv 1 \bmod 4$ and $h = 2$ otherwise. Denote $G = \{1, \sigma\}$.

$\underline{H^q(G, \mathcal{O}_L)}$: Let $\alpha = x + y\sqrt{d} \in \mathcal{O}_L$. Then $\mathrm{N}_G\, \alpha = 2x$, so ${}_{\mathrm{N}_G}\mathcal{O}_L = \mathbb{Z}\sqrt{d}$ and $\mathrm{N}_G\, \mathcal{O}_L = h\mathbb{Z}$. Furthermore

$$I_G\mathcal{O}_L = \{\sigma\alpha - \alpha \mid \alpha \in \mathcal{O}_L\} = \{2y\sqrt{d} \mid y \in \tfrac{h}{2}\mathbb{Z}\} = h\mathbb{Z}\sqrt{d}$$

and $\mathcal{O}_L^G = \mathcal{O}_L \cap \mathbb{Q} = \mathbb{Z}$, hence

$$H^{-1}(G, \mathcal{O}_L) \cong {}_{\mathrm{N}_G}\mathcal{O}_L / I_G\mathcal{O}_L \cong \begin{cases} 0 \\ \mathbb{Z}/2\mathbb{Z} \end{cases}, \quad H^0(G, \mathcal{O}_L) \cong \mathcal{O}_L^G / \mathrm{N}_G\, \mathcal{O}_L \cong \begin{cases} 0 & \text{if } d \equiv 1 \bmod 4 \\ \mathbb{Z}/2\mathbb{Z} & \text{otherwise.} \end{cases}$$

$\underline{H^q(G, \mu_L)}$: On μ_L we have that N_G is trivial, $I_G\mathcal{O}_L^\times = \mu_L^2$ and $(\mu_L^\times)^G = \mu_L^\times \cap \mathbb{Q} = \{\pm 1\}$. Hence $H^{-1}(G, \mu_L^\times) \cong \mathbb{Z}/2\mathbb{Z} \cong H^0(G, \mu_L^\times)$.

$\underline{H^q(G, \mathcal{O}_L^\times)}$: If $d < 0$, then $\mathcal{O}_L^\times = \mu_L$. So let $d > 0$, $\varepsilon \in \mathcal{O}_L^\times$ a fundamental unit, and assume $\mathrm{N}(\varepsilon) = 1$, so that $\mathrm{N}_G = 2$. We have $I_G\mathcal{O}_L = \langle \varepsilon^{\sigma - 1} \rangle = \langle \varepsilon^{-2} \rangle$ and $(\mathcal{O}_L^\times)^G = \{\pm 1\}$ as before, leading to $H^{-1}(G, \mathcal{O}_L^\times) \cong (\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z})/(0 \times 2\mathbb{Z}) \cong (\mathbb{Z}/2\mathbb{Z})^2$ and $H^0(G, \mathcal{O}_L^\times) \cong \mathbb{Z}/2\mathbb{Z}$.

Now suppose $\mathrm{N}(\varepsilon) = -1$. Then $\mathrm{N}_G\, \mathcal{O}_L^\times = \{\pm 1\} = (\mathcal{O}_L^\times)^G$, so $H^0(G, \mathcal{O}_L^\times) = 0$ and ${}_{\mathrm{N}_G}\mathcal{O}_L^\times = \langle \pm\varepsilon^2 \rangle$ while $I_G\mathcal{O}_L^\times = \langle -\varepsilon^{-2} \rangle$, hence $H^{-1}(G, \mathcal{O}_L^\times) \cong \mathbb{Z}/2\mathbb{Z}$.

$\underline{H^q(G, I_L)}$: We have $I_L \cong \bigoplus_{\mathfrak{p}} \mathbb{Z} \cong \bigoplus_p \bigoplus_{\mathfrak{p}|p} \mathbb{Z}$, where p denotes primes in $\mathbb{Z}$ and $\mathfrak{p}$ primes in L. Since the last decomposition respects the G-module structure, we have $H^q(G, I_L) \cong \bigoplus_p H^q(G, \bigoplus_{\mathfrak{p}|p} \mathbb{Z})$. If p does not split in L, then the action of G is trivial on $I_p := \bigoplus_{\mathfrak{p}|p} \mathbb{Z} = \mathfrak{p}\mathbb{Z}$, so $H^{-1}(G, I_p) = 0$ and $H^0(G, I_p) = \mathbb{Z}/2\mathbb{Z}$. If p splits, then σ permutes the factors of $I_p = \mathbb{Z} \oplus \mathbb{Z}$, so that ${}_{\mathrm{N}_G}I_p = \langle (1, -1) \rangle$, $\mathrm{N}_G\, I_p = \langle (1, 1) \rangle = I_p^G$ as well as

$$I_G I_p = \{(1 - \sigma)(x, y) \mid x, y \in \mathbb{Z}\} = \{(x - y, y - x) \mid x, y \in \mathbb{Z}\} = \langle (1, -1) \rangle,$$

hence $H^{-1}(G, I_L) = 0 = H^0(G, I_L)$. Putting things together, we find

$$H^{-1}(G, I_L) = 0, \qquad H^0(G, I_L) \cong \bigoplus_{p \text{ non-split in } L} \mathbb{Z}/2\mathbb{Z}.$$

Exercise 3

By Hilbert's Theorem 90 (see ANT1, Sheet 14, Exercise 4) we have

$$H^{-1}(G, L^\times) \cong {}_{\mathrm{N}_G}L^\times / I_G L^\times = \{a \in L^\times \mid \mathrm{N}_{L/K}(a) = 1\} / \{\tfrac{\sigma(b)}{b} \mid b \in L^\times\} = 0.$$

Exercise 4

(a) Since $(p, m) = 1$ we know that f is separable. Let $\alpha_1, \dots, \alpha_m$ be the pairwise different roots of f. Then $(\frac{\alpha_i}{\alpha_1})^m = \frac{a}{a} = 1$, so the $\frac{\alpha_i}{\alpha_1}$ are pairwise different m-th roots of unity. This immediately implies $\mu_m = \{\frac{\alpha_i}{\alpha_1} \mid i = 1, \dots, m\} \subseteq L$.

(b) It is clear that x is a well-defined map, we have to show that it is a crossed homomorphism. So let $\sigma, \tau \in G$. Then

$$x(\sigma\tau) = \frac{\sigma\tau(\omega)}{\omega} = \frac{\sigma\tau(\omega)}{\sigma(\omega)}\frac{\sigma(\omega)}{\omega} = \sigma(x(\tau))x(\sigma),$$

so $\overline{x}$ is an element of $H^1(G, \mu_m)$. Now let $\omega' \in L$ be another element with $\omega'^m = a$, then $\omega' = \zeta\omega$ for some $\zeta \in \mu_m$, and $\frac{\sigma(\omega')}{\omega'} = \frac{\sigma(\eta)}{\eta}\frac{\sigma(\omega)}{\omega}$ with $\sigma \mapsto \frac{\sigma(\eta)}{\eta}$ being a 1-coboundary. This shows that the class $\overline{x} \in H^1(G, \mu_m)$ is independent of the choice of ω.

Exercise 1

(a) $A^G \subseteq A^U$ is clear, so it remains to show $N_G A \subseteq N_U A$. Let $\sigma_1, \ldots, \sigma_k$ be a system of representatives for $U \setminus G$, and set $\mathrm{N}_{U\setminus G} = \sum_{i=1}^k \sigma_i$. Then $N_G = \sum_{g \in G} g = \sum_{i=1}^k \sum_{u \in U} u\sigma_i = \mathrm{N}_U \, \mathrm{N}_{U\setminus G}$, which immediately implies the claim.

(b) Let $\overline{c} \in H^0(G, C)$. Let $\overline{b} \in H^0(G, B)$ with $jb = c$, and $\overline{a} \in H^1(G, A)$ with $i_* a = \partial b$, so that $\delta(\overline{c}) = \overline{a}$. Now clearly $i(\mathrm{res}_1(a)) = \mathrm{res}_1(\partial b) = \partial_U b$, so $\delta(\mathrm{Res}_0(\overline{c})) = \mathrm{Res}_1(\overline{a}) = \mathrm{Res}_1(\delta(\overline{c}))$

Exercise 2

Consider the diagram

$$\begin{tikzcd}
U^{\mathrm{ab}} \arrow[r, "\cong"] \arrow[dd, "\varphi"] & I_U/I_U^2 = H^{-1}(U, I_U) \arrow[d, "i_*"] & H^{-2}(U, \mathbb{Z}) \arrow[l, "\delta"'] \arrow[dl, "\delta"] \arrow[dd, "\mathrm{Kor}_{-2}"] \\
 & H^{-1}(U, I_G) \arrow[d, "\mathrm{Kor}_{-1}"] & \\
G^{\mathrm{ab}} \arrow[r, "\cong"] & I_G/I_G^2 = H^{-1}(G, I_G) & H^{-2}(G, \mathbb{Z}) \arrow[l, "\delta"']
\end{tikzcd}$$

Here, the trapezoid on the right commutes by definition of corestriction with the short exact sequence $0 \to I_G \to \mathbb{Z}[G] \to \mathbb{Z} \to 0$, and the commutativity of the triangle follows from naturality of the long exact cohomology sequence. The big rectangle commutes by definition of φ and Exercise 4.2, hence the left square also commutes. Hence for $g \in U$, the image of $\varphi(gU')$ is given by

$$\mathrm{Kor}_{-1}(i_*(g - 1 + I_U^2)) = \mathrm{Kor}_{-1}(g - 1 + I_U I_G) = g - 1 + I_G^2,$$

which maps to gG' in G^{ab}. Hence $\varphi(gU') = gG'$.

Exercise 3

By definition, $\mathrm{Kor}_0 : H^0(U, L^\times) = F^\times / \mathrm{N}_{L/F} L^\times \to K^\times / \mathrm{N}_{L/K} L^\times = H^0(G, L^\times)$ is given by $\alpha \, \mathrm{N}_{L/F} L^\times \mapsto \mathrm{N}_{G/U}(\alpha) \, \mathrm{N}_{L/K} L^\times$, where $\mathrm{N}_{G/U} \in \mathbb{Z}[G]$ is the sum of a system of representatives of G/U. Let $\sigma_1, \ldots, \sigma_k \in \mathrm{Gal}(L/K)$ be such a system, then the $\sigma_i|_F$ are $|G/U| = [F : K]$ pairwise different morphisms $F \to \overline{K}$, hence for $\alpha \in F^\times$ we have $\mathrm{N}_{F/K}(\alpha) = \prod_{i=1}^k \sigma_i(\alpha) = \mathrm{N}_{G/U}(\alpha)$ as required. The argument for the trace is exactly the same, reading everyting additively.

Exercise 4

Given $x : G^q \to A$, let $f^*x : (G')^q \to f^*A$, $(\sigma_1, \ldots, \sigma_q) \mapsto x(f(\sigma_1), \ldots, f(\sigma_q))$, we want to show that this induces a morphism $f^* : H^q(G, A) \to H^q(G', f^*A)$, i.e. we have to show that the diagram

$$\begin{tikzcd}
A_q \arrow[r, "\partial"] \arrow[d, "f^*"] & A_{q+1} \arrow[d, "f^*"] \\
(f^*A)_q \arrow[r, "\partial"] & (f^*A)_{q+1}
\end{tikzcd}$$

commutes for $q \geq 0$. But this is clear from the explicit formulas for ∂. Now composition with $g_* : H^q(G', f^*A) \to H^q(G', A')$ gives the desired morphism $(f, g)^*$.

If $f : H \hookrightarrow G$ is the inclusion of a subgroup, then for $x \in A_q$, $f^*x = x|_{H^q}$ by definition, which induces the restriction. Now let $H \subseteq G$ be a normal subgroup, $f : G \to G/H$ the projection and $g : A^H \to A$ the inclusion. It is clear that f and g are compatible. By definition, (f, g) is induced by the map of cochains $(x : (G/H)^q \to A^H) \mapsto (G^q \to A, \ \sigma \mapsto g(x(f(\sigma))))$, which is just like inflation is defined.

Exercise 1

(a) Let $\overline{c} \in H^q(G, C)$, $c \in C_q$. Let $b \in B_q$ with $j_* b = c$ and $a \in A_{q+1}$ with $i_* a = \partial b$, so that $\delta(\overline{c}) = \overline{a}$ and $\operatorname{res}_{q+1}(\delta(\overline{c})) = \overline{a|_{g^q}}$. On the other hand, we have $j_* b|_{g^q} = (j_* b)|_{g^q} = c|_{g^q}$ and $i_* a|_{g^q} = (\partial b)|_{g^q} = \partial(b|_{g^q})$, which means $\delta(\operatorname{res}_q \overline{c}) = \overline{a|_{g^q}}$.

(b) Follows directly from the uniqueness of restrictions in last week's exercises.

Exercise 2

$N_G a_1(\sigma) = \sum_{\tau \in G} \tau a_1(\sigma) = \sum_{\tau \in G} a_1(\sigma\tau) - a_1(\tau) = 0$ shows $a_1(\sigma) \in {}_{N_G}A$, i.e. $a_1(\sigma) + I_G A \in H^{-1}(G, A)$. Now let a_1 be a 1-coboundary, that is $a_1(\sigma) = \sigma x - x$ for some $x \in A$. This says exactly that $a_1(\sigma) \in I_G A$, i.e. $a_1(\sigma) + I_G A$ is trivial. Hence $a_1(\sigma) + I_G A$ only depends on the class $\overline{a}_1 \in H^1(G, A)$.

Exercise 3

Let $\rho \in G$. Using the 2-cocycle condition, we have

$$\rho a_0 = \sum_{\tau \in G} \rho a_2(\tau, \sigma) = \sum_{\tau \in G} a_2(\rho\tau, \sigma) + a_2(\rho, \tau) - a_2(\rho, \tau\sigma) = \sum_{\tau \in G} a_2(\rho\tau, \sigma) = a_0,$$

hence $a_0 \in A^G$ and $\overline{a}_0 \in H^0(G, A)$ is well-defined. Now let a_2 be a 2-coboundary, that is $a_2(\sigma, \tau) = \sigma y(\tau) - y(\sigma\tau) + y(\sigma)$ for some $y : G \to A$; we have to show that a_0 is a norm. Plugging in this formula yields

$$a_0 = \sum_{\tau \in G} a_2(\tau, \sigma) = \sum_{\tau \in G} \tau y(\sigma) - y(\tau\sigma) + y(\tau) = \sum_{\tau \in G} y(\sigma) = N_G y(\sigma).$$

Exercise 4

(a) Let $u' : G \to E$ be another section of π, then $\pi(u'_x u_x^{-1}) = x x^{-1} = 1$, so by exactness $u'_x = u_x a'$ for some $a' \in A$. Then $u'_x a (u'_x)^{-1} = u_x a' a (a')^{-1} u_x^{-1} = u_x a u_x^{-1}$ since A is abelian.

(b) Follows directly from equation (2).

(c) As above, $\pi(u_x u_y u_{xy}^{-1}) = 1$, so there exists some $f(x, y) \in A$ with $u_x u_y = f(x, y) u_{xy}$.

(d) Let $x, y, z \in G$. One computes

$$\begin{aligned} f(x,y) f(xy,z) u_{xyz} &= f(x,y) u_{xy} u_z = (u_x u_y) u_z = u_x (u_y u_z) \\ &= u_x f(y,z) u_{yz} = x f(y,z) u_x u_{yz} = x f(y,z) f(x,yz) u_{xyz} \end{aligned}$$

and cancelling u_{xyz} gives exactly the 2-cocycle condition.

(e) Let $f' \in Z^2(G, A)$ be the cocycle obtained from u'. Then $u' \cdot u^{-1}$ has values in A, i.e. is a 1-cochain, and

$$\begin{aligned} f(x,y) \partial_2(u' u^{-1})(x,y) &= f(x,y) x(u'_y u_y^{-1}) u_{xy} (u'_{xy})^{-1} u'_x u_x^{-1} \\ &= u_x u'_y u_y^{-1} u_x^{-1} f(x,y) u_{xy} (u'_{xy})^{-1} u'_x u_x^{-1} \\ &= u_x u'_y (u'_{xy})^{-1} u'_x u_x^{-1} = u_x (u'_x)^{-1} f'(x,y) u'_x u_x^{-1} \\ &= x x^{-1} f'(x,y) = f'(x,y). \end{aligned}$$

Exercise 1

(a) Since $\mathrm{Aut}(C_2) = 1$ it is clear that G can only act trivially on C_2. Now since G is cyclic, $H^2(G, C_2) \cong H^0(G, C_2) \cong C_2/N_G C_2 = C_2$, so there are two extensions of C_2 by C_2. The two given ones are clearly non-isomorphic, so they represent exactly the classes of such extensions.

(b) $\mathbb{C}^\times \leq W$ is clear, and since $zj = j\overline{z}$, so is $j\mathbb{C}^\times \leq W$. It remains to show that W is closed under products, but this and the homomorphism property of π follow immediately from $z \cdot z' = zz'$, $z \cdot jz' = j(\overline{z}z')$ and $jz \cdot jz' = -\overline{z}z'$ for $z, z' \in \mathbb{C}^\times$.

(c) Again, G is cyclic, so $H^2(G, \mathbb{C}^\times) \cong H^0(G, \mathbb{C}^\times) \cong \mathbb{R}^\times / \mathrm{N}_G\, \mathbb{C}^\times = \mathbb{R}^\times/(\mathbb{R}^\times)^2 \cong \{\pm 1\}$, where the last isomorphism is induced by the sign function. The two given extensions are clearly non-isomorphic, we have to verify that they are extensions compatible with the G-module structure on $\mathbb{C}^\times$. In the first case this is clear since it is split. For the second, choose a section $1 \mapsto 1$, $\tau \mapsto j$ of π, then $\tau z = jzj^{-1}$ for $z \in \mathbb{C}^\times$ is exactly the required condition (2) from last sheet.

Exercise 2

Consider the coaugmentation sequence $A \xrightarrow{i} A \otimes \mathbb{Z}[G] \xrightarrow{j} A^1$. Then $i_* \overline{a}_2 \in H^2(G, A \otimes \mathbb{Z}[G]) = 0$, so $ia_2 = \partial a_1$ for some 1-cochain $a_1 \in (A \otimes \mathbb{Z}[G])_1$. By construction it follows that $\delta(j_* \overline{a}_1) = \overline{a}_2$ and hence

$$\overline{a}_2 \cup \overline{\sigma} = \delta(j_* \overline{a}_1) \cup \overline{\sigma} = \delta(j_* \overline{a}_1 \cup \overline{\sigma}) = \delta(\overline{ja_1(\sigma)})$$

by definition of the cup-product and a formula from the lecture. Now again using the construction of δ, let $ia_0 = \partial a_1(\sigma)$, so that $\delta(\overline{ja_1(\sigma)}) = \overline{a}_0$. Then , the relation $ia_2 = \partial a_1$ means $ia_2(\tau, \sigma) = \tau a_1(\sigma) - a_1(\tau\sigma) + a_1(\tau)$, so

$$i(a_0) = \partial a_1(\sigma) = \sum_{\tau \in G} \tau a_1(\sigma) = \sum_{\tau \in G} ia_2(\tau, \sigma) + \sum_{\tau \in G} a_1(\tau\sigma) - \sum_{\tau \in G} a_1(\tau) = i\left(\sum_{\tau \in G} a_2(\tau, \sigma)\right)$$

and injectivity of i yields the claim.

Exercise 3

We have $(N \circ D)(a) = N_G(\sigma - 1)a = (1-1)N_G a = 0$ and $(D \circ N)(a) = (\sigma - 1)N_G a = N_G a - N_G a = 0$. Now $\ker N = {}_{N_G}A$ and $\operatorname{im} N = N_G A$ are clear. $a \in \ker D$ is equivalent to $\sigma a = a$, which is equivalent to $a \in A^G$ since σ generates G. Finally, $\operatorname{im} D = (\sigma - 1)A = I_G A$ since $\sigma - 1 \mid \sigma^k - 1$ in $\mathbb{Z}[G]$. Hence $(\ker D : \operatorname{im} N) = |H^0(G, A)|$ and $(\ker N : \operatorname{im} D) = |H^{-1}(G, A)| = |H^1(G, A)|$, so that $h(A) = q_{D,N}(A)$.

Exercise 4

$\ker 0 = A = A^G$ and $\operatorname{im} g = nA = N_G A$ since G acts trivially. Hence $(\ker f : \operatorname{im} g) = |H^0(G, A)|$. Similarly, $\ker g = {}_{N_G}A$ and $\operatorname{im} 0 = 0 = I_G A$, so $(\ker g : \operatorname{im} f) = |H^{-1}(G, A)|$ and $h(A) = q_{0,n}(A)$.

Exercise 5

Consider the commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & A \arrow[r] \arrow[d, "n\cdot"] & B \arrow[r] \arrow[d, "n\cdot"] & C \arrow[r] \arrow[d, "n\cdot"] & 0 \\
0 \arrow[r] & A \arrow[r] & B \arrow[r] & C \arrow[r] & 0
\end{tikzcd}$$

The snake lemma yields the exact sequence

$$0 \to A[n] \to B[n] \to C[n] \to A/nA \to B/nB \to C/nC \to 0.$$

By definition, $q_{0,n}(X) = |X/nX|/|X[n]|$, so if two of $q_{0,n}(A), q_{0,n}(B), q_{0,n}(C)$ exist, then four of the terms in the sequence are finite, and hence the other two sit between two finite groups, therefore are finite themselves. Thus all groups are finite, and the exact sequence yields

$$|A[n]| \cdot |C[n]| \cdot |B/nB| = |B[n]| \cdot |A/nA| \cdot |C/nC|,$$

which after rearranging is just $q_{0,n}(B) = q_{0,n}(A)q_{0,n}(C)$.

Exercise 6

From the given diagram the snake lemma gives a short exact sequence

$$0 \to \ker f/(g(A) \cap \ker f) \to A/g(A) \to f(A)/fg(A) \to 0.$$

Now $g(A) \cap \ker f \cong \ker fg/\ker g$ via g, and $f(A)/fg(A) \cong \frac{A/fg(A)}{A/f(A)}$, hence

$$(A : g(A)) \cdot |\ker fg| \cdot (A : f(A)) = |\ker f| \cdot |\ker g| \cdot (A : fg(A)),$$

a priori as cardinalities. We show that if two of $q_{0,f}, q_{0,g}, q_{0,fg}$ exist, then so does the third; then the desired equality follows immediately.

If $q_{0,f}$ and $q_{0,g}$ exist, then $|\ker fg| \leq |\ker f| \cdot |\ker g|$ is finite, and if $q_{0,fg}$ exists, then $(A : f(A)), (A : g(A)) \leq (A : fg(A))$ are finite. Then from the above equality, the last term has to be finite as well.

Exercise 7

[Neukirch I.6.9]

Exercise 1

Let $a : G \to L^\times$ be a 1-cocycle. By independence of characters, we can find $c \in L$ with $b := \sum_{\tau\in G} a(\tau)\tau(c) \neq 0$. Then

$$\sigma(b) = \sum_{\tau\in G} \sigma(a(\tau))\sigma\tau(c) = a(\sigma)^{-1}a(\sigma\tau)\sigma\tau(c) = a(\sigma)^{-1}b,$$

so $a(\sigma) = \frac{\sigma(b^{-1})}{b^{-1}}$ is a 1-coboundary.

Exercise 2

(1)⇒(2): Denote the action by φ and let $a \in A$. Since A is discrete, $\varphi^{-1}(a) \cap (G \times \{a\}) = \{(x, a) \mid \varphi(x, a) = a\} = G_a$ is open.

(2)⇒(3): $a \in A^{G_a}$ for all $a \in A$, so $A = \bigcup_a A^{G_a} \subseteq \bigcup_U A^U$.

(3)⇒(1): Let $b \in A$, we want to show that $\varphi^{-1}(b)$ is open. Let $(g, a) \in \varphi^{-1}(b)$, and pick an open subgroup $U \subseteq G$ with $a \in A^U$. For $u \in U$, $\varphi(gu, a) = b$, so $gU \times \{a\} \subseteq \varphi^{-1}(b)$ and $\varphi^{-1}(b)$ is open.

Exercise 3

By infinite Galois theory, $G_{\mathbb{Q}}^{ab} \cong \mathrm{Gal}(\overline{\mathbb{Q}}^{G'_{\mathbb{Q}}}/\mathbb{Q})$. We have $\overline{\mathbb{Q}}^{G'_{\mathbb{Q}}} = \mathbb{Q}^{\mathrm{ab}}$, since for $\sigma, \tau \in G_{\mathbb{Q}}$, $[\sigma, \tau]$ acts as the identity on $\overline{\mathbb{Q}}^{G'_{\mathbb{Q}}}$, so $\overline{\mathbb{Q}}^{G'_{\mathbb{Q}}}$ is abelian, and conversely, if $a \in \overline{\mathbb{Q}} \setminus \overline{\mathbb{Q}}^{G'_{\mathbb{Q}}}$, and $a \in K$ with $K/\mathbb{Q}$ finite galois, then by the same argument K cannot be abelian.

By the Kronecker-Weber theorem, $\mathbb{Q}^{\mathrm{ab}} = \varinjlim_{K/\mathbb{Q}\text{ fin. ab.}} K = \varinjlim_n \mathbb{Q}(\zeta_n) =: \mathbb{Q}(\zeta_\infty)$. Hence

$$G_{\mathbb{Q}}^{\mathrm{ab}} \cong \mathrm{Gal}(\mathbb{Q}(\zeta_\infty)/\mathbb{Q}) \cong \varprojlim_n \mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong \varprojlim_n (\mathbb{Z}/n\mathbb{Z})^\times \cong (\varprojlim_n \mathbb{Z}/n\mathbb{Z})^\times = \widehat{\mathbb{Z}}^\times.$$

Exercise 4

(a) Let $\varphi : \widehat{\mathbb{Z}} \to \mathbb{Z}/n\mathbb{Z}$ be the natural projection. Clearly $n \in \ker\varphi$, so φ induces a morphism $\overline{\varphi} : \widehat{\mathbb{Z}}/n\widehat{\mathbb{Z}} \to \mathbb{Z}/n\mathbb{Z}$. Converely, by the universal property of limits, the family of projections $\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ induces a morphism $\psi : \mathbb{Z} \to \widehat{\mathbb{Z}}$, which again induces a morphism $\overline{\psi} : \mathbb{Z}/n\mathbb{Z} \to \widehat{\mathbb{Z}}/n\widehat{\mathbb{Z}}$.

Now, $\varphi \circ \psi = \varphi$ by construction, so $\overline{\varphi} \circ \overline{\psi} = \mathrm{id}_{\mathbb{Z}/n\mathbb{Z}}$. Conversely, let $x \in \widehat{\mathbb{Z}}$. Then $\overline{\varphi}(\overline{x}) = x_n$, say represented by $y \in \mathbb{Z}$, hence $\overline{\psi} \circ \overline{\varphi}(\overline{x}) = \psi(y) + n\widehat{\mathbb{Z}}$. So for $\overline{\psi} \circ \overline{\varphi} = \mathrm{id}_{\widehat{\mathbb{Z}}/n\widehat{\mathbb{Z}}}$ we have to show $x - \psi(y) \in n\widehat{\mathbb{Z}}$, which follows directly from $\psi(y)_n = x_n$.

Indeed, for any $x \in \widehat{\mathbb{Z}}$ and $n \in \mathbb{N}$, one has $x + n\widehat{\mathbb{Z}} = \{y \in \widehat{\mathbb{Z}} \mid x_n = y_n\}$: "$\subseteq$" is clear, so suppose $y \in \widehat{\mathbb{Z}}$ with $x_n = y_n$. For $k \in \mathbb{N}$, $y_{kn} \equiv x_n \bmod n$, so $y_{kn} = nz_k + x_n$ for some z_k. Then $z = (z_k)_k \in \widehat{\mathbb{Z}}$ satisfies $x + nz = y$

(b) Follows directly from Galois theory with $\widehat{\mathbb{Z}} \cong \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q) =: G$, $1 \mapsto \varphi$ and $\overline{\mathbb{F}}_q^{\varphi} = \mathbb{F}_q = \overline{\mathbb{F}}_q^{G}$.